

Solutions to JEE Advanced – 1 | JEE 2023 | Paper-1 (Gen 1 & 2)

PHYSICS

- 1.(B) Take x axis along the incline and y direction perpendicular to it.

Velocity of rain relative to the man is perpendicular to the incline in this case (i.e., along the umbrella stick. This keeps canopy perpendicular to the rainfall and provides maximum safety).

$$\begin{aligned}\vec{V}_{rm} &= \vec{V}_r - \vec{V}_m \\ &= (V_x \hat{i} + V_y \hat{j}) - V_0 \hat{i} = (V_x - V_0) \hat{i} + V_y \hat{j}\end{aligned}$$

Since $\vec{V}_{rm}$ has no x component

$$\therefore V_x = V_0$$

When the man is walking up, $\vec{V}_{rm}$ is directed vertically downward.

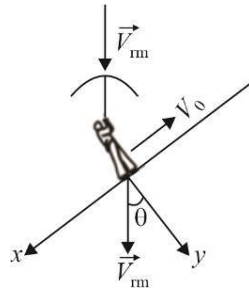
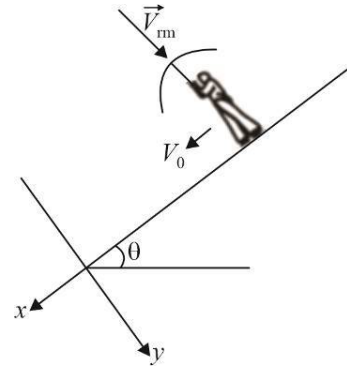
$$\begin{aligned}\vec{V}_{rm} &= (V_x \hat{i} + V_y \hat{j}) - (-V_0 \hat{i}) \\ &= V_0 \hat{i} + V_y \hat{j} + V_0 \hat{i} \\ &= 2V_0 \hat{i} + V_y \hat{j}\end{aligned}$$

From diagram

$$\tan \theta = \frac{2V_0}{V_y}$$

$$\frac{3}{4} = \frac{2V_0}{V_y} \Rightarrow V_y = \frac{8V_0}{3}$$

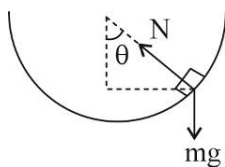
$$\therefore V_r = \sqrt{V_x^2 + V_y^2} = \frac{\sqrt{73}}{3} V_0$$



- 2.(B) $\cos \theta = \frac{1}{2} \Rightarrow \theta = 60^\circ$

$$N \cos \theta = mg$$

$$N \sin \theta = mw^2(R \sin \theta)$$



$$\tan \theta = \frac{w^2 R \sin \theta}{g}$$

$$\sqrt{3} = \frac{w^2 R}{g} \Rightarrow w = \sqrt{\frac{2g}{R}}$$

- 3.(D) The elevator moves up with constant acceleration, hence y-t graph must be a parabola.

$$\text{Let } y = kt^2$$

$$\text{At } t = 2, y = 4$$

$$\therefore k = 1 \quad \therefore y = t^2 \quad \therefore \frac{dy}{dt} = 2t = 4 \text{ m/s (at } t = 2)$$

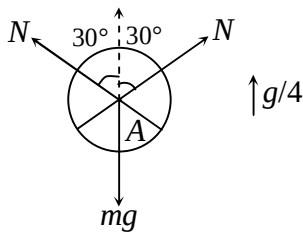
$$\frac{d^2y}{dt^2} = 2.0 \text{ m/s}^2$$

In the reference frame of the elevator the acceleration of bolt is 12 m/s^2 and its initial velocity is zero. Time required

for a displacement of 1.5 m in this frame is $y = \frac{1}{2} \times 12 \times t^2$

$$1.5 = \frac{1}{2} \times 12 \times t^2 \Rightarrow t = 0.5 \text{ s} \quad \therefore \text{ Bolt hits the floor at } t = 2.5 \text{ s}$$

- 4.(B) For sphere A, $N\sqrt{3} = mg + \frac{mg}{4}$, $N = \frac{5mg}{4\sqrt{3}}$



- 5.(A) $\vec{v}_B = 8\hat{j} + 2t\hat{k}$

$$\vec{v}_C = \vec{v}_0 = v_x\hat{i} + v_y\hat{j} + v_z\hat{k}$$

$$\vec{r}_B = 8t\hat{j} + t^2\hat{k}$$

$$\vec{r}_C = v_x t\hat{i} + v_y t\hat{j} + v_z t\hat{k}$$

$$\text{At 4 sec, } \vec{r}_B = \vec{r}_C \Rightarrow v_x = 0, v_y = 8 \text{ m/s and } v_z = 4 \text{ m/s} \quad \therefore \vec{v}_0 = (8\hat{j} + 4\hat{k}) \text{ m/s}$$

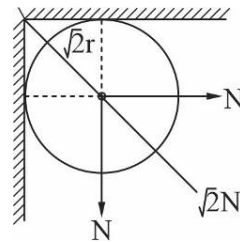
- 6.(B) $a = -s$, $v \frac{dv}{ds} = -s$, $\int_{v_0}^0 v dv = -\int_0^s s ds$, $\frac{v_0^2}{2} = \frac{s^2}{2} \Rightarrow s = v_0$

- 7.(AC) The thread makes an angle of $\sin^{-1}\left(\frac{\sqrt{2}r}{2r}\right) = 45^\circ$ with the vertical.

If N normal force by each wall on the cylinder

$$\sqrt{2}N = \frac{T}{\sqrt{2}} \quad \text{and} \quad Mg = \frac{T}{\sqrt{2}}$$

$$\Rightarrow T = \sqrt{2}Mg \quad \text{and} \quad N = \frac{Mg}{\sqrt{2}}$$



8.(AB) For completing circle $u = \sqrt{4gl}$

Speed of bob when rod becomes horizontal can be found by energy conservation

$$\frac{1}{2}m(4gl) = \frac{1}{2}mv^2 + mgl \Rightarrow v = \sqrt{2gl}$$

Radial acceleration, $a_r = \frac{v^2}{l} = 2g$

Tangential acceleration, $a_t = \frac{mg}{m} = g$ $\therefore$ Net acceleration $= \sqrt{a_r^2 + a_t^2} = \sqrt{5}g$

At the topmost point, velocity is zero

So, force by rod on bob is mg radially outwards. So, C and D are incorrect

9.(BD) If the height of the pan at an instant is l and the boy transfers a small mass m of the sand into it, he performs a work $= mgl$ (against gravity).

This work done increases the gravitational potential energy of the sand mass which loses a part of it in compressing the spring.

Therefore, total work done by the boy will be finally found as gravitational potential energy of the sand plus the elastic potential energy of the spring.

As per the question.

$$K \frac{L_0}{2} = Mg \quad \dots (i)$$

$$\text{Work done } W = Mg \frac{L_0}{2} + \frac{1}{2} K \left(\frac{L_0}{2} \right)^2$$

$$= Mg \frac{L_0}{2} + \frac{1}{8} KL_0^2 = Mg \frac{L_0}{2} + \frac{1}{4} MgL_0 \quad [\text{using (1)}]$$

$$= \frac{3}{4} MgL_0$$

10.(ABC) Friction force between the block and the plank $= f$

$$\text{Acceleration of the block } a = \frac{f}{M}$$

If the slipping stops in time 't'

$$V = 0 + at$$

$$V = \frac{ft}{M} \quad \dots (1)$$

For the plank external force $F = \text{friction } (f)$

Power of the external force $P = FV$

Work done by external force in time 't'

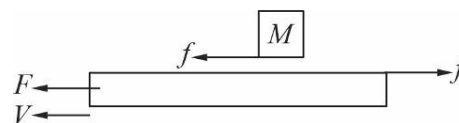
$$W = Pt = FVt = ftV = MV^2 [\because ft = MV \text{ from (1)}]$$

The work done by the external force can be interpreted as

$W = \text{Heat produced} + \text{gain in KE of the block}$

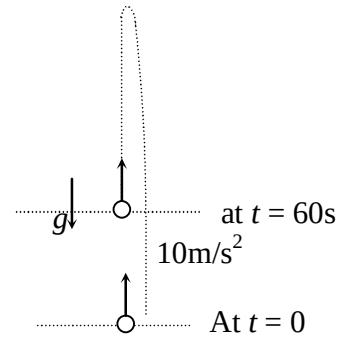
$$\therefore \text{Heat} = MV^2 - \frac{1}{2} MV^2 = \frac{1}{2} MV^2$$

$$\frac{1}{2} MV^2 \text{ amount of KE. This energy has been lost as heat.}$$



$$11.(BD) \text{ Total height} = \frac{1}{2} \times 10 \times (60)^2 + \frac{(10 \times 60)^2}{2 \times 10} = 36000 \text{ m} = 36 \text{ km}$$

$$\begin{aligned} \text{Total time} &= 60 + 60 + \sqrt{\frac{2 \times 36000}{10}} \\ &= 60s + 60s + 60\sqrt{2}s \\ &= (120 + 60\sqrt{2})s \end{aligned}$$



$$12.(AC) \text{ Force on the particle, } F = -\frac{dU}{dr} = -3mkr^2$$

The $-ve$ sign indicates attractive force.

For the motion to be circular

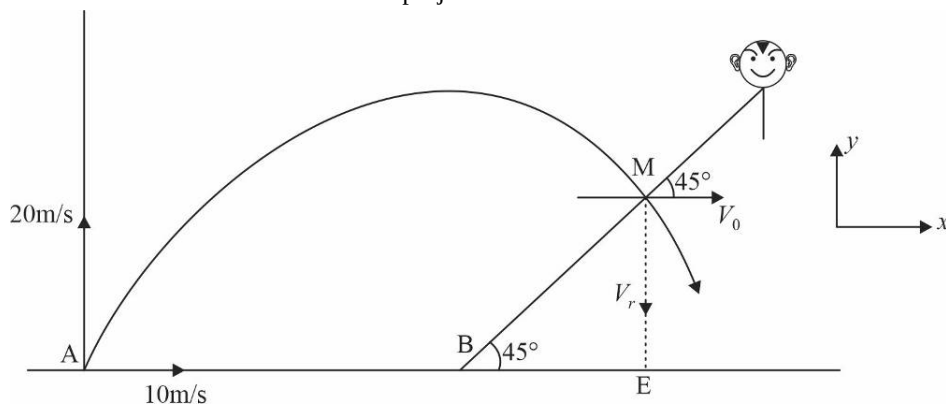
$$\frac{mv^2}{r} = 3mkr^2 \Rightarrow v = r\sqrt{3kr} \quad \dots\dots(i)$$

$$\text{For } r = a_0 \quad \therefore KE = \frac{1}{2}mv^2 = \frac{3}{2}kma_0^3 \quad ; \quad \text{Time period, } T = \frac{2\pi(2a_0)}{v} = \frac{2\pi 2a_0}{2a_0\sqrt{3k 2a_0}} = \frac{2\pi}{\sqrt{6ka_0}}$$

13.(15) to 14.(15)

In the reference frame of the wedge, initial velocity of the particle is $\vec{V} = 10\hat{i} + 20\hat{j} \text{ m/s}$

Hence, the motion in this frame is that of a usual projectile



This projectile hits the wedge normally, means its velocity at the instant of hit is inclined at 45° to the horizontal

$$\therefore \tan 45^\circ = \left| \frac{V_y}{V_x} \right| \Rightarrow v_y = -10 \quad ; \quad -10 = 20 - 10t \quad \therefore t = 3 \text{ sec}$$

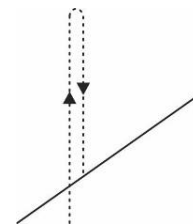
Height of projectile at this instant

$$ME = 20 \times 3 - \frac{1}{2} \times 10 \times 3^2 = 15 \text{ m} ;$$

$$\frac{ME}{BE} = \tan 45^\circ \Rightarrow BE = ME = 15 \text{ m} \quad \text{and} \quad AE = 10 \times 3 = 30 \text{ m}$$

$$AB = 30 - 15 = 15 \text{ m}$$

B will be 15m to the left of A.



- 15.(20) Till the time B doesn't start slipping, there will be no friction between A and B
Maximum static friction between B and ground is

$$f_{1m} = (0.2)(300) = 60N$$

Slipping starts when $F = f_{1m}$

$$3t = 60 \Rightarrow t = 20 \text{ s}$$

- 16.(70) Maximum static friction between

A and B is $f_{2m} = (0.5)(10) = 50$

Maximum possible acceleration of A is $a_{\max} = \frac{f_{2m}}{m_A} = \frac{50}{10} = 5 \text{ m/s}^2$

Slipping between A and B starts when acceleration of system reacts this value

$$F - 60 = 30(5)$$

$$F = 60 + 150$$

$$3t = 210 \Rightarrow t = 70 \text{ sec}$$

- 17.(10) 18.(19)

Since wall is smooth time of flight remains unchanged

$$T = \frac{2u_y}{g} = \frac{2 \times 100}{10} = 20 \text{ s}$$

The particle reaches the wall at $t = \frac{600}{60} \text{ s} = 10 \text{ s}$

At $t = 10 \text{ s}$, the particle is at its maximum height. Hence $t = 10 \text{ s}$ and $\theta = 0^\circ$

Let Q be the point of impact on the wall.

So its X - co-ordinate $= 80 \times 10 = 800 \text{ m}$

$$\text{Y - co-ordinate} = \frac{u_y^2}{2g} = \frac{(100)^2}{2 \times 10} = 500 \text{ m}$$

$$\text{Z - co-ordinate} = 0$$

Since only velocity component along line of impact changes so, just after collision with wall.

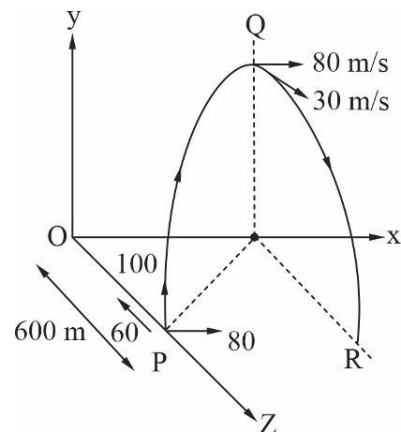
$$\vec{V}_Q = 80\hat{i} + e \times 60\hat{k} = (80\hat{i} + 30\hat{k}) \text{ m/s}$$

And, time to reach the XZ - plane is 10s from Q, so, for R,

$$\text{X - co-ordinate} = 80 \times 20 = 1600 \text{ m}$$

$$\text{Y - co-ordinate} = 0$$

$$\text{Z - co-ordinate} = 30 \times 10 = 300 \text{ m}$$



CHEMISTRY

- 1.(B) The given radial wave function belongs to 3s-orbital.

I.E. of the H-atom for its electron present in 3s energy level is $\frac{E_0(1)^2}{3^2}$

$$\text{i.e. } \frac{E_0}{9} = \frac{2.18 \times 10^{-18}}{9} = 2.42 \times 10^{-19} \text{ J}$$

$$\text{Energy of the incident photon, } E_{\text{photon}} = \frac{hc}{\lambda} = 5 \times 10^{-19} \text{ J}$$

Now using the photoelectric equation $E_{\text{photon}} = (\text{I.E.})_{\text{atom}} + (\text{K.E.})_{e^-}$

$$\Rightarrow (\text{K.E.})_{e^-} = (5 - 2.42) \times 10^{-19} = 2.58 \times 10^{-19} \text{ J}$$

2.(C) (A) $[\text{OH}^-] = \frac{1000 \text{ md}}{1000 + \text{mM}^\circ} = \frac{1000 \times 2 \times 1.2}{1000 + 2(40)} = \frac{2400}{1080}$

$$\Rightarrow [\text{OH}^-]_A = 2.22 \text{ M}$$

(B) $[\text{OH}^-]_B = 1.15 \times 2 = 2.30 \text{ M}$

(C) $[\text{OH}^-]_C = \frac{10 \times 10}{40} = 2.5 \text{ M}$

(D) $[\text{OH}^-]_D = 2 \left[\frac{8 \times 10}{74} \right] = \frac{160}{74} = 2.16 \text{ M}$

- 3.(D) As HCl is a polar bond then $d_{\text{H-Cl}} < (r_{\text{H}} + r_{\text{Cl}})$.

$$\text{Here } r_{\text{H}} = \frac{74}{2} = 37 \text{ pm, } r_{\text{Cl}} = \frac{110}{2} = 55 \text{ pm}$$

$\therefore d_{\text{H-Cl}} = \text{Bond length of HCl} < 92 \text{ pm}$; the H-Cl bond strength will be definitely greater than bond strength of Cl_2 , thus $B.E_{\text{HCl}} > 350 \text{ kJ/mole}$.

- 4.(A) Group 17th elements are called as salt producers i.e. halogens

- 5.(A) For a real gas obeying $\left(P + \frac{a}{V^2}\right)(V - b) = RT$, gas molecules have both attractive and repulsive interaction thus graph of option (A) is most appropriate.

- 6.(C) According to quantum mechanical model, the position of first maxima in the $4\pi r^2 \psi^2(r)$ v/s $\frac{r}{a_0}$ curves provide information about potential energy.

Distance of 1st maxima from the nucleus : $B < C < A$

Potential energy : $B < C < A$

Penetration effect : $B > C > A$

Number of spherical nodes : $\underset{(2)}{B} > \underset{(1)}{C} > \underset{(0)}{A}$

- 7.(BCD)

Based upon electronic configurations in neutral and +1 state; orders of I.E. are

1st I.E. : $\text{N} > \text{O}, \text{Be} > \text{B}$

2nd I.E. : $\text{O} > \text{N}, \text{O} > \text{F}$

8.(CD) According to M.O.T; s-p mixing is NOT operative in diatomic species if atomic number of the participating element is ≥ 8

Compounds	$\Rightarrow$	KCN	CaC_2	KO_2	Na_2O_2
Anion	$\Rightarrow$	CN^-	C_2^{2-}	O_2^-	O_2^{2-}
s-p mixing	$\Rightarrow$	✓	✓	✗	✗

9.(ABC) $Z_m = 1$ at the conditions specified in option A, B and C. At critical conditions $Z_m = \frac{3}{8}$

10.(ABCD) In context to structure of an atom, all of the given statements are correct.

11.(ACD)

0.1 M, 100 ml of H_2SO_4

$\Rightarrow$ 10 m moles of H_2SO_4 (or) 20 milli-eql. H_2SO_4

It can be neutralize completely by :

(A) 0.02 gm-eql. (or) 20 milli-eql. of Ba(OH)_2

(B) 20 milli eql. of Na_2CO_3

(C) 20 m moles of $\text{KOH} \equiv$ 20 milli-eql. of KOH

(D) 0.8 gm of $\text{NaOH} \equiv$ 20 milli-eql. of NaOH

12.(ABCD)

All of the given statements are incorrect w.r.t. kinetic theory of gases.

13.(20) In presence of phenolphthalein as the indicator.

$$n_{\text{HCl}} = n_{\text{Na}_2\text{CO}_3} + n_{\text{NaOH}} = 0.1 + 0.1 = 0.2 \text{ mole}$$

$$\text{Molarity of the HCl solution} = \frac{10 \times d}{M^\circ} = \frac{10(29.2)(1.25)}{36.5} = 10 \text{ M}$$

$$\Rightarrow 10 \times V_{(\text{L})} = 0.2 \Rightarrow V_{(\text{L})} = 0.02 \text{ L} = 20 \text{ ml} \quad \therefore Z = 20$$

14.(30) On using methyl orange as the indicator

$$10(V)_{(\text{L})} = 2n_{\text{Na}_2\text{CO}_3} + n_{\text{NaOH}} = 0.3$$

$$(V)_{(\text{L})} = 0.03(\text{L}) = 30 \text{ ml} \quad \therefore W = 30$$

15.(8) State S1 : $\left. \begin{array}{l} \text{Spherically symmetrical} \Rightarrow s\text{-orbital, } \ell = 0 \\ \text{One radial node} \Rightarrow n - \ell - 1 = 1, n - 1 = 1, n = 2 \end{array} \right\} 2s$

State S2 : $\left. \begin{array}{l} \text{One angular node} \Rightarrow \ell = 1 \\ \text{One radial node} \Rightarrow n - \ell - 1 = 1, n - 2 = 1, n = 3 \end{array} \right\} 3p$

$$\text{For H-atom; } E_1 = -E_0 \frac{Z^2}{n^2} = -E_0$$

$$\text{For } n = 1, Z = 1, E_0 = 13.6 \text{ eV / atom}$$

For transition of e^- in He^+

$$\Delta E = E_0 Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] = 13.6 [2]^2 \left[\frac{1}{2^2} - \frac{1}{3^2} \right] = 7.55$$

16.(9) State S2 : Belongs to $n = 3$

$$\text{Number of degenerate orbitals} = n^2 = 3^2 = 9$$

$$17.(500) \frac{r_X}{r_Y} = \sqrt{\frac{M_Y^\circ}{M_X^\circ}} = \frac{10 \text{ mm} / 5}{10 \text{ mm} / 10} = 2$$

$$\Rightarrow \frac{M_Y^\circ}{M_X^\circ} = 4 \text{ or } M_X^\circ = 20 \text{ g mol}^{-1}$$

$$\text{Now } V_{(\text{rms})} \text{ of } X(g) = \sqrt{\frac{3RT}{M^\circ}}$$

$$\text{Put } R = 8.33, T = 200, M^\circ = 20 \times 10^{-3} \text{ kg / mole}$$

$$V_{(\text{rms})} = V = \sqrt{\frac{3 \times 8.33 \times 200}{20 \times 10^{-3}}} = \sqrt{\frac{25 \times 200}{20 \times 10^{-3}}} \Rightarrow V = 5 \times 10^2 \text{ m / sec} = 500 \text{ m / sec}$$

18.(1200)

Let at temperature T (Kelvin)

$$V_{\text{MP}}[Y] = V_{\text{RMS}}[X] \text{ at } [200 \text{ K}]$$

$$\Rightarrow \sqrt{\frac{2RT}{M_Y^\circ}} = \sqrt{\frac{3RT'}{M_X^\circ}} \Rightarrow \frac{2T}{M_Y^\circ} = \frac{3[200]}{M_X^\circ}$$

$$\text{Or } T = \frac{3[200]}{2 M_X^\circ} [M_Y^\circ] = 3 \times 100 \times 4 = 1200 \text{ K}$$

MATHEMATICS

1.(B) $17^{1995} = (10 + 7)^{1995}$

$17^{1995} - 7^{1995}$ is a multiple of 10

$11^{1995} = (10 + 1)^{1995}$.

2.(B) Com fig: we see that

$$OC_1^2 = a_1^2 + b_1^2 \quad \dots (i)$$

$$OC_2^2 = a_2^2 + b_2^2 \quad \dots (ii)$$

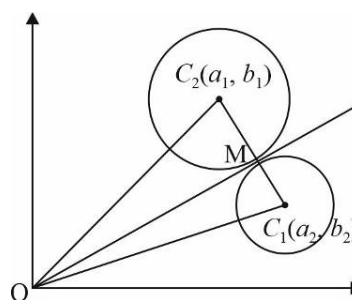
Also

$$(OM)^2 = (OC_1)^2 - (C_1M)^2 = (OC_2)^2 - (C_2M)^2$$

$$\Rightarrow a_1^2 + b_1^2 - r_1^2 = a_2^2 + b_2^2 - r_2^2$$

$$\Rightarrow (a_1^2 + b_1^2) - (a_2^2 + b_2^2) = r_1^2 - r_2^2$$

$$\Rightarrow (a_1^2 - a_2^2) + (b_1^2 - b_2^2) = (r_1^2 - r_2^2)$$



3.(A) The equation of any lines passing through the given point $P(3, 4)$ and making an angle $\frac{\pi}{6}$ with x-axis is

$$\frac{x-3}{\cos 30^\circ} = \frac{y-4}{\sin 30^\circ} = r \quad (\text{say}) \quad \dots \dots \dots (i)$$

Where r represents the distance of any point Q on this line from the given point $P(3, 4)$

The coordinates (x, y) of any point Q on line (i) are

$$(3 + r \cos 30^\circ, 4 + r \sin 30^\circ) \text{ i.e., } \left(3 + \frac{r\sqrt{3}}{2}, 4 + \frac{r}{2} \right)$$

If the point lies on the line $12x + 5y + 10 = 0$, then

$$12 \left(3 + \frac{r\sqrt{3}}{2} \right) + 5 \left(4 + \frac{r}{2} \right) + 10 = 0 \Rightarrow \frac{132}{12\sqrt{3} + 5}$$

4.(B) Let the coordinates of the point P which divides the line joining $(1, 0)$ and $(2 \cos \theta, 2 \sin \theta)$ in the ratio $2 : 3$ be (h, k) . Then

$$h = \frac{4 \cos \theta + 3}{5} \text{ and } k = \frac{4 \sin \theta}{5} \Rightarrow \cos \theta = \frac{5h-3}{4} \text{ and } \sin \theta = \frac{5k}{4}$$

$$\Rightarrow \left(\frac{5h-3}{4} \right)^2 + \left(\frac{5k}{4} \right)^2 = 1$$

$(5h-3)^2 + (5k)^2 = 16$ Therefore locus of (h, k) is $(5x-3)^2 + (5y)^2 = 16$ which is a circle

5.(A) $z^6 - 1 = (z-1)(1+z+z^2+\dots+z^5)$

$$\Rightarrow z^6 - 1 = (z-1)(z-\alpha_1)(z-\alpha_2)\dots\dots(z-\alpha_5)$$

Put $x = 2$

$$\frac{2^6 - 1}{2 - 1}$$

6.(C) $z - 2 = re^{i\alpha}$

$$\therefore (x - 2) + iy = r(\cos \theta + i \sin \alpha)$$

$$r^2 = (x - 2)^2 + y^2 \text{ and } \tan \alpha = \frac{y}{x - 2}$$

The expression $\frac{z - 2}{e^{i\theta}} + \frac{e^{i\theta}}{z - 2} = re^{i(\alpha - \theta)} + \frac{1}{r}e^{-i(\alpha - \theta)}$

Which is given as real

$$\therefore r \sin(\alpha - \theta) - \frac{1}{r} \sin(\alpha - \theta) = 0$$

$$\Rightarrow r - \frac{1}{r} = 0 \Rightarrow r^2 = 1 \Rightarrow (x - 2)^2 + y^2 = 1$$

7.(ABD) We have $(1 + \sqrt{2})^n = x_n + y_n \sqrt{2} \dots (1)$

$$(1 - \sqrt{2})^n = x_n - y_n \sqrt{2} \dots (2)$$

From (1) and (2), we get

$$(1 + \sqrt{2})^n (1 - \sqrt{2})^n = x_n^2 - 2y_n^2 \Rightarrow (-1)^n = x_n^2 - 2y_n^2$$

Next, $x_{n+1} + y_{n+1} \sqrt{2} = (1 + \sqrt{2})^{n+1} = (1 + \sqrt{2})(x_n + \sqrt{2}y_n) = (x_n + 2y_n) + \sqrt{2}(x_n + y_n)$

Thus, $x_{n+1} = x_n + 2y_n$ and $y_{n+1} = x_n + y_n$

8.(AB) $\therefore (1 + x + x^2)^n = a_0 + a_1 x + a_2 x^2 + \dots + a_{2n} x^{2n}$

Putting $x = i (i = \sqrt{-1})$. Then, we get

$$(1 + i + i^2)^n = (a_0 - a_2 + a_4 - a_6 + \dots) + i(a_1 - a_3 + a_5 - a_7 + \dots)$$

$$\Rightarrow i^n = (a_0 - a_2 + a_4 - a_6 + \dots) + i(a_1 - a_3 + a_5 - a_7 + \dots)$$

If n is odd, then $\operatorname{Re}(i^n) = 0$

$$\Rightarrow a_0 - a_2 + a_4 - a_6 + \dots = 0$$

If n is even, then $\operatorname{Im}(i^n) = 0$

$$\Rightarrow a_1 - a_3 + a_5 - a_7 + \dots = 0$$

9.(BC) Centre of the circle must be $(1, 2)$

Centre of second circle $= (-1, 0)$, radius $= \sqrt{2}$.

The distance between centres $= 2\sqrt{2}$ which must be either $r + \sqrt{2}$ or $r - \sqrt{2} \Rightarrow r = \sqrt{2}$ or $3\sqrt{2}$

10.(BC) As the required circle touches the y-axis at (0, 2), let its equation be

$$(x - \alpha)^2 + (y - 2)^2 = \alpha^2 \text{ or } x^2 + y^2 - 2\alpha x - 4y + 4 = 0.$$

This circle meets the x-axis at the points where $x^2 - 2\alpha x + 4 = 0$

which gives two values of x, say x_1, x_2 , such that $x_1 + x_2 = 2\alpha$ and $x_1 x_2 = 4$.

Now, we are given that $|x_1 - x_2| = 3$.

$$\therefore (x_1 + x_2)^2 - 4x_1 x_2 = (x_1 - x_2)^2 = 9 \Rightarrow 4\alpha^2 - 16 = 9$$

$$\Rightarrow \alpha^2 = \frac{25}{4} \Rightarrow \alpha = \pm \frac{5}{2}$$

Hence the equation of the required circle is $x^2 + y^2 \pm 5x - 4y + 4 = 0$

11.(BC) $4x + 3y = 10$ is equation of common tangent at $P(1, 2)$.

$$\text{Equation of normal at } P \text{ is } y - 2 = \frac{3}{4}(x - 1)$$

$$\text{As slope of } 4x + 3y = 10 \text{ is } -\frac{4}{3} \Rightarrow 3x - 4y + 5 = 0$$

$$\text{Centre } (\alpha, \beta) \text{ lies on it. } \therefore 3\alpha - 4\beta + 5 = 0$$

$$\Rightarrow \frac{3\alpha + 5}{4} = \beta \quad (2)$$

$$\text{Now radius} = 5 \Rightarrow (CP)^2 = 25 \Rightarrow (\alpha - 1)^2 + \frac{9}{16}(\alpha - 1)^2 = 25$$

$$\Rightarrow (\alpha - 1)^2 = 16$$

$$\Rightarrow \alpha - 1 = \pm 4 \Rightarrow \alpha = 5, -3$$

Now put this in (2), $\beta = 5, -1$. Thus $(\alpha, \beta) = (5, 5), (-3, -1)$.

$$\text{Equation of circle is } (x - 5)^2 + (y - 5)^2 = 5^2 \text{ and } (x + 3)^2 + (y + 1)^2 = 5^2$$

$$\Rightarrow x^2 + y^2 - 10x - 10y + 25 = 0 \text{ and } x^2 + y^2 + 6x + 2y - 15 = 0.$$

12.(AB)

The x-coordinates of the points of intersection of the line $y = mx + c$ and the circle $x^2 + y^2 = r^2$ are given by

$$x^2 + (mx + c)^2 = r^2$$

$$\text{or } (1 + m^2)x^2 + 2mcx + c^2 - r^2 = 0 \quad \dots (i)$$

Which, being quadratic in x, gives two values of x and hence two points of intersection. These points will be real and distinct if the discriminant of (i) is positive i.e.,

$$4m^2c^2 - 4(1 + m^2)(c^2 - r^2) > 0 \Rightarrow c^2 < r^2(1 + m^2) \Rightarrow r\sqrt{1 + m^2} < c < r\sqrt{1 + m^2}$$

$$13.(6) \left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^6 = \sum_{r=0}^9 {}^9C_r \left(\frac{3}{2}x^2\right)^{9-r} \left(-\frac{1}{3x}\right)^r$$

For the term that is independent of x,

$$\text{We must have } 18 - 2r - r = 0$$

$$\Rightarrow r = 6$$

$$\text{Required term} = {}^9C_6 \left(\frac{3}{2} x^2 \right)^3 \left(-\frac{1}{3x} \right)^6 = {}^9C_6 \left(\frac{1}{6} \right)^3$$

14.(0) Let $(r+1)^{\text{th}}$ term of $\left(\frac{5}{x^2} + x^4 \right)^n$ be independent of x , we have

$$T_{r+1} = {}^nC_r \left(\frac{5}{x^2} \right)^{n-r} (x^4)^r = {}^nC_r 5^{n-r} x^{6r-2n}$$

For this term to be independent of x , $6r - 2n = 0$ or $n = 3r$

$\therefore$ Each of 18, 21, 27, 99 is divisible by 3.

15.(0) $1 + \alpha^{10} + \alpha^{20} + \alpha^{30} + \dots + \alpha^{190} = 0$

As 10 is not an integral multiple of n .

16.(1) $n = 5, \alpha^5 = 1$

and $1 + \alpha + \alpha^2 + \alpha^3 + \alpha^4 = 0$

$$(1 + \alpha)(1 + \alpha^2)(1 + \alpha^3)(1 + \alpha^4)$$

$$= (1 + \alpha + \alpha^2 + \alpha^3)(1 + \alpha^3 + \alpha^4 + \alpha^7)$$

$$= (-\alpha^4)(1 + \alpha^2 + \alpha^3 + \alpha^4) = (-\alpha^4)(-\alpha) = \alpha^5 = 1$$

17.(60) Area of rectangle PQRS = $12 \times 5 = 60$ sq. units

18.(30) Now $C_1 : (x-4)(x+8) + (y-3)(y+2) = 0$

or $x^2 + y^2 + 4x - 6y - 23 = 0$

centre $(-2, 3)$ & Radius = 6

Equation of tangent through P to C_1 is $y + 2 = m_1 (x + 8)$

$$\Rightarrow \left| \frac{-6m_1 - 5}{\sqrt{1 + m_1^2}} \right| = 6 \Rightarrow m_1 = \frac{11}{60} \quad \therefore |m_1| + |m_2| = \frac{11}{30}$$

PHYSICS

1.(ACD) $mg \cos \theta - N = \frac{mv^2}{R}$

Condition for leaving contact : $N = 0$

$$\therefore v^2 = Rg \cos \theta$$

Energy conservation gives

$$\frac{1}{2}mv^2 = mg(h + R - R \cos \theta)$$

$$\frac{1}{2}m(Rg \cos \theta) = mg(h + R - R \cos \theta)$$

$$\frac{1}{2}R \cos \theta + R \cos \theta = h + R$$

$$\frac{3}{2}R \cos \theta = h + R$$

$$\text{For } h = \frac{R}{4}; \frac{3}{2}R \cos \theta = \frac{5R}{4} \Rightarrow \cos \theta = \frac{5}{6}$$

$$\text{For } h = \frac{R}{3}; \frac{3}{2}R \cos \theta = \frac{4R}{3} \Rightarrow \cos \theta = \frac{8}{9}$$

$$\text{For } h = 0; \frac{3}{2}R \cos \theta = R \Rightarrow \cos \theta = \frac{2}{3}$$

$$\text{For } \theta = 0; \frac{3}{2}R = h + R \Rightarrow h = \frac{R}{2}$$

2.(BCD) (AB) $a = g \sin \theta - \mu g \cos \theta$

$$= 10 \times \frac{3}{5} - 0.5 \times 10 \times \frac{4}{5} = 2 \text{ m/s}^2$$

(C) $F_{\min} = mg \sin \theta - \mu mg \cos \theta = 10 \text{ N}$

(D) $mg \sin \theta = \mu(mg \cos \theta + F)$

$$30 = 0.5(40 + F) \Rightarrow F = 20 \text{ N}$$

3.(AD) The components of two vectors along a direction perpendicular to their resultant must be equal and opposite.

$$\text{So, } |\vec{A}| \sin 30^\circ = |\vec{B}| \sin 45^\circ \Rightarrow |\vec{A}| = \sqrt{2} |\vec{B}|$$

Also, the components of the two vectors along the direction of their resultant must add up to their resultant.

$$\text{Therefore, } |\vec{R}| = |\vec{A}| \cos 30^\circ + |\vec{B}| \cos 45^\circ \Rightarrow |\vec{R}| = \frac{\sqrt{3}}{2} |\vec{A}| + \frac{1}{\sqrt{2}} |\vec{B}|$$

4.(ABC) $\Delta u = (u^2 + u^2 + 2u^2 \cos 120^\circ)^{\frac{1}{2}}$

$$\Delta u = u = 60 \text{ ms}^{-1}$$

$$\text{Average acceleration} = \frac{\Delta u}{t} = \frac{u}{\frac{r\theta}{60}} = 11.5 \text{ ms}^{-2}$$

$$\text{Instantaneous acceleration} = \frac{u^2}{r} = 12 \text{ ms}^{-2}$$

5.(BD) Compression is maximum when velocity of block is zero.

Applying energy conservation

$$mg(h+x) = \frac{1}{2}kx^2 \Rightarrow 10(0.4+x) = 50x^2 \Rightarrow 5x^2 - x - 0.4 = 0$$

$$x = \frac{1 \pm \sqrt{1+8}}{10} = \frac{1+3}{10} = 0.4 \text{ m}$$

$$\text{Velocity of block is maximum when } mg = kx \Rightarrow x = \frac{10}{100} = 0.1 \text{ m}$$

$$\text{Applying energy conservation } mg(h+x) = \frac{1}{2}kx^2 + \frac{1}{2}mv^2$$

$$10(.4+0.1) = \frac{1}{2}(100)(.1)^2 + \frac{1}{2}(1)v^2$$

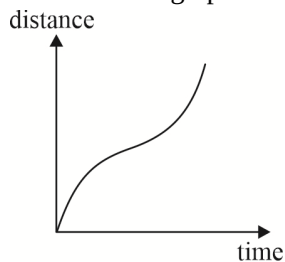
$$5 = 0.5 + \frac{1}{2}v^2 \Rightarrow v^2 = 9 \Rightarrow v = 3 \text{ m/s}$$

6.(BC) Acceleration is constant and -ve

$\therefore$ Velocity-time graph is a straight line with -ve slope

Displacement-time graph is a parabola opening downwards

Correct distance-time graph is



7.(C) $\frac{1}{2}mv_0^2 = mgh + (\lambda h)g \frac{h}{2}$

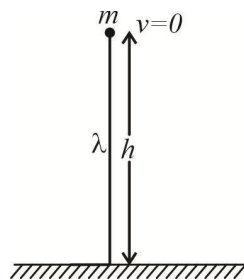
$$\frac{1}{2}(5)(400 \times 6) = 5(10)h + (0.5)\frac{h^2}{2}(10)$$

$$6000 = 50h + 2.5h^2$$

$$h^2 + 20h - 2400 = 0$$

$$(h+60)(h-40) = 0$$

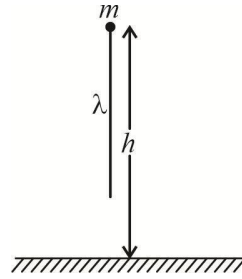
$$h = 40 \text{ m}$$



- 8.(D) As the length of chain is less than $40m$, it will be completely lifted off

$$\begin{aligned}\frac{1}{2}mv_0^2 &= mgh + \lambda(30)g(h-15) \\ \frac{1}{2}(5)(400 \times 6) &= 50h + 150(h-15) \\ 6000 &= 200h - 2250 \\ h &= \frac{8250}{200} = 41.25m\end{aligned}$$

Height of C.M of chain $= h - 15 = 26.25m$



- 9.(B) C.M remains at rest

$$\therefore \Delta x_{cm} = 0$$

$$m_1 \Delta x_1 + m \Delta x_2 = 0$$

Let balloon moves down by y

$$\therefore M(-y) + m(\ell - y) = 0 \quad \Rightarrow \quad y = \frac{m\ell}{m+M}$$

- 10.(C) C.M. moves with constant velocity

$$\therefore \Delta x_{cm} = v_0 t$$

Where t is the time taken by man to reach the balloon. $t = \frac{l}{u}$

Let balloon moves up by y

$$\frac{My + m(l+y)}{M+m} = V_0 \frac{l}{u}$$

$$(M+m)y = (M+m)V_0 \frac{l}{u} - m\ell ; \quad y = V_0 \frac{l}{u} - \frac{m}{m+M}l$$

- 11.(C) $a = 2 - 0.5t$

$$\frac{dv}{dt} = 2 - 0.5t$$

$$\int_0^v dv = \int_0^t (2 - 0.5t) dt$$

$$v = 2t - 0.25t^2$$

$$v = 0 \text{ at } t = 0 \text{ and } t = 8$$

$$\frac{dx}{dt} = 2t - 0.25t^2$$

$$\int_0^x dx = \int_0^t (2t - 0.25t^2) dt ; \quad d = \left[t^2 - \frac{t^3}{12} \right]_0^8 = 64 - \frac{512}{12} = 21.33m$$

- 12.(B) Velocity is maximum when $a = 0$ i.e. at $t = 4$

$$v = 2(4) - (0.25)(16) = 4 \text{ m/s}$$

13.(2.81) Speed just before reaching B is given by energy conservation

$$mg(5) = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2(10)(5)} = 10\text{m/s}$$

After collision at B, velocity perpendicular to the incline becomes zero while velocity along the incline remains unchanged

$$\therefore v_1 = v \cos 30^\circ = 5\sqrt{3} \text{ m/s}$$

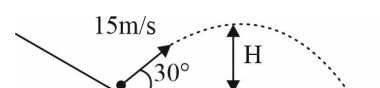
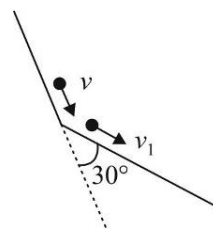
Velocity upon reaching C can be found by applying energy conservation again.

$$mg(7.5) = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 \Rightarrow v_2^2 = v_1^2 + 2g(7.5) = 75 + 150 = 225$$

$$\therefore v_2 = 15 \text{ m/s}$$

As the collision with floor is elastic, the particle bounces with same speed making same angle with the floor.

$$H = \frac{(15)^2 \sin^2 30^\circ}{2g} = \frac{225}{80} = 2.81 \text{ m}$$



14.(600) From A(t=0) to B(t=6)

$$a = \frac{F-f}{m} = 10^{-5} = 5 \text{ m/s}^2 ; s_1 = \frac{1}{2}at^2 = \frac{1}{2}(5)(36) = 90 \text{ m}$$

$$\text{Velocity at B} = at = 5(6) = 30 \text{ m/s}$$

After B, the direction of F is reversed.

As velocity is in forward direction, friction acts backwards till block comes to instantaneous rest at C.
From B to C

$$a = -\frac{(F+f)}{m} = -(10+5) = -15 \text{ m/s}^2$$

$$u = 30 \text{ m/s} ; v^2 = u^2 + 2as_2 ; 0 = (30)^2 + 2(-15)s_2 \Rightarrow s_2 = 30 \text{ m}$$

While returning from C to A, F acts towards left but friction acts towards right (as velocity is towards left)

$$s = -(s_1 + s_2) = -120 \text{ m} ; s = -\frac{(F-f)}{m} = -5\text{m/s}^2$$

$$v^2 = u^2 + 2as = 0 + 2(-120)(-5) ; v^2 = 1200 \Rightarrow v = 20\sqrt{3}$$

15.(2.25) For A and C

$$v_A \cos 53^\circ = v_C \sin 53^\circ$$

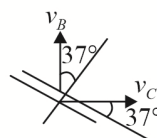
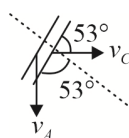
$$v_C = v_A \cot 53^\circ$$

$$= 4\left(\frac{3}{4}\right) = 3 \text{ m/s}$$

For B and C

$$v_B \cos 37^\circ = v_C \sin 37^\circ$$

$$v_B = v_C \tan 37^\circ = 3\left(\frac{3}{4}\right) = 2.25 \text{ cm/s}$$



16.(0.80) Time taken to cross in shortest time = $\frac{d}{5}$

Time taken to cross over shortest distance = $\frac{d}{\sqrt{5^2 - 3^2}} = \frac{d}{4}$ $\therefore$ Ratio = $\frac{\frac{d}{5}}{\frac{d}{4}} = \frac{4}{5} = 0.8$

17.(8) Let the acceleration of the blocks be a_1 , a_2 and a_3 respectively, all assumed upwards

Let the mass of A be m

Then, $2T - mg = ma_1$

$T - 3g = 3a_2$

$T - 2g = 2a_3$

Therefore, $a_1 = \frac{2T}{m} - g$; $a_2 = \frac{T}{3} - g$; $a_3 = \frac{T}{2} - g$

But we know that $a_2 + a_3 = -2a_1$

$\Rightarrow \left(\frac{T}{3} - g\right) + \left(\frac{T}{2} - g\right) = -2\left(\frac{2T}{m} - g\right)$

$\Rightarrow T = \frac{4g}{\frac{5}{6} + \frac{4}{m}} = \frac{24mg}{(5m + 24)}$

But, for the block B to accelerate upwards, $a_2 > 0$

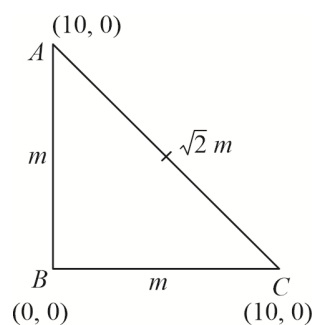
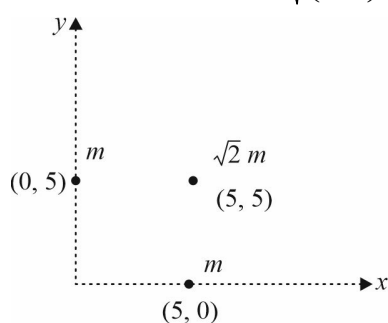
$\Rightarrow \frac{T}{3} - g > 0 \Rightarrow T > 3g \Rightarrow \frac{24mg}{(5m + 24)} > 3g \Rightarrow m > 8$

18.(5) To find C.M, the three rods can be replaced by point masses lying on their C.M.

$X_{cm} = \frac{m(5) + m(0) + \sqrt{2} m(5)}{m + m + \sqrt{2}m} = \frac{5}{\sqrt{2}} \text{ cm}$

Similarly, $r_{cm} = \frac{5}{\sqrt{2}} \text{ cm}$

$\therefore$ Distance from origin = $\sqrt{\left(\frac{5}{\sqrt{2}}\right)^2 + \left(\frac{5}{\sqrt{2}}\right)^2} = 5 \text{ cm}$



CHEMISTRY

1.(ABC)

Statement A, B and C are correct. Ground state electronic configuration of ${}_{24}\text{Cr}$ is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$

2.(CD) In set (A) : Only XeF_6 is polar

In set (B) : NF_3 and IF_3 are polar

In set (C) and (D), all compounds have permanent dipole.

3.(ABCD)

All of the given statements are correct w.r.t. the Maxwell-Boltzmann distribution of molecular speeds.

4.(ABCD)

If an atomic orbital can have upto 3 electrons then electronic configuration will be

${}_4\text{Be} : 1s^3 2s^1$, first alkali metal

${}_{31}\text{Ga} : 1s^3 2s^3 2p^9 3s^3 3p^9 4s^3 3d^1$, first transition metal

${}_{15}\text{P} : 1s^3 2s^3 2p^9$, second noble gas

${}_{16}\text{S} : 1s^3 2s^3 2p^9 3s^1$, second alkali metal

5.(BCD) Both B_2^- and C_2^- are paramagnetic; statements of option B, C and D are correct according to M.O.T.6.(BC) $\psi_{200} \Rightarrow 2s$, $\psi_{210} \Rightarrow 2p_z$, $\psi_{300} \Rightarrow 3s$

$\psi_{310} \Rightarrow 3p_z$, $\psi_{320} \Rightarrow 3d_{z^2}$, $\psi_{100} \Rightarrow 1s$

$\psi_{400} \Rightarrow 4s$, $\psi_{420} \Rightarrow 4d_{z^2}$

As in a H-atom $E_{2p} = E_{2s}$ thus that transition will be non-radiative.

7.(D) Stationary dipole-dipole interaction : $E \propto \frac{1}{r^3}$

Hydrogen bonding : $E \propto \frac{1}{r^x}$, $x \neq 6$

Ion-dipole interaction : $E \propto \frac{1}{r^2}$

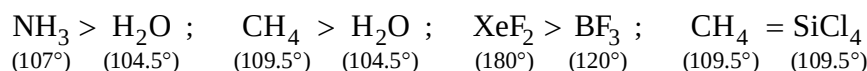
Induced dipole-induced dipole interaction : $E \propto \frac{1}{r^6}$

8.(A) H_2O and $\text{O}_2\text{N}-\text{C}_6\text{H}_4-\text{OH}$ shows strong intermolecular H-bonding thus their boiling points are greater than CH_3OH (showing weak intermolecular H-bonding) and $\text{O}_2\text{N}-\text{C}_6\text{H}_3(\text{OH})-\text{NO}_2$ (showing intramolecular H-bonding) respectively.

9.(D) According to VSEPR theory the most stable/geometry of the given molecules are

XeF_4 : Square planar; SF_4 : See saw; XeF_2 : Linear; ClF_3 : Bent T-shape

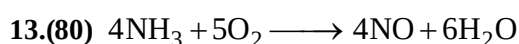
10.(A) Correct orders of bond angle



11.(B) $T_B = \frac{a}{Rb} = -3^\circ\text{C} = 273 - 3 = 270\text{ K}$

$$T_C = \frac{8}{27} \left[\frac{a}{Rb} \right] = \frac{8}{27} [T_B] = \frac{8}{27} [270] = 80\text{ K}$$

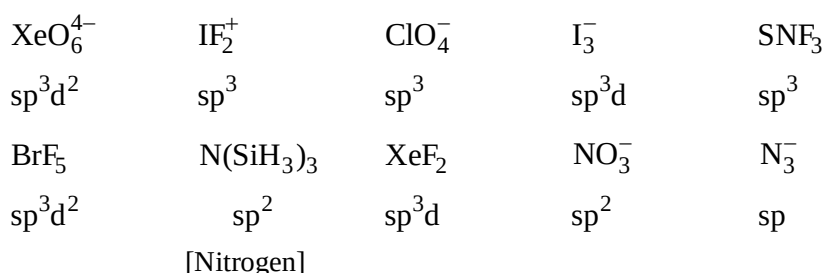
12.(C) $P_C = \frac{a}{27b^2} = \frac{2.7}{27[0.1]^2} = 10\text{ atm}$



$\Rightarrow$ 1 mole of NH_3 can give 0.8 mole of NO [80% yield]

$\therefore$ 1.7 kg i.e. 100 moles of NH_3 can give 80 moles of NO(g).

14.(4) Hybrid states of the central atoms.



15.(6) As per the data given

Most stable positive valency of A = 1

Most stable positive valency of B = 3

Most stable positive valency of C = 2 $\therefore$ Sum of $1 + 3 + 2 = 6$

16.(0.50) As in 4 out of 5 experiments, volume of NaOH used is 16 mL thus that will be the most appropriate value of volume of NaOH needed for complete neutralisation of the oxalic acid. By law of chemical equivalence.

Milli eq. of oxalic acid = milli eq. of NaOH

$$10 \times 0.4 \times 2 = 16[\text{NaOH}](1), [\text{NaOH}] = 0.5\text{ M}$$

17.(8) Molar ratio of gases after "n" steps

$$\frac{n_{\text{H}_2}}{n_{\text{D}_2}} = \frac{P_{\text{H}_2}^\circ}{P_{\text{D}_2}^\circ} \left[\frac{M_{\text{D}_2}^\circ}{M_{\text{H}_2}^\circ} \right]^{n/2} \Rightarrow \frac{16}{1} = \frac{1}{1} \left[\frac{4}{2} \right]^{n/2} \Rightarrow 16 = 2^{n/2} \Rightarrow (16)^2 = 2^n \therefore n = 8$$

18.(0.50) In single electron atomic species

$$\lambda = 2\pi a_0 \left[\frac{n}{z} \right] \quad a_0 = 0.53 \text{ \AA} = 0.053 \text{ nm}$$

$$= 2 \times 3.14 \times 0.053 \times \frac{3}{2} = 0.49926 \text{ nm}$$

$$\lambda = 0.5 \text{ nm}$$

MATHEMATICS

1.(AB) The coordinates of A and B are (0, 12) and (8, 0) respectively.

The equation of the perpendicular bisector of AB is

$$y - 6 = \frac{2}{3}(x - 4) \text{ or } 2x - 3y + 10 = 0 \quad \dots\dots\dots(i)$$

Equation of a line passing through (0, 1) and parallel to x-axis is $y = -1$

This meets (i) at C, Therefore the coordinates of C are $\left(-\frac{13}{2}, -1\right)$

$$\text{Hence the area of the triangle ABC is } \Delta = \frac{1}{2} \begin{vmatrix} 0 & 12 & 1 \\ 8 & 0 & 1 \\ -\frac{13}{2} & -1 & 1 \end{vmatrix} = 91 \text{ sq. units}$$

2.(AC) The given lines being parallel tangents to a circle, the diameter of the circle is equal to the distance between

$$\text{these lines, so that the required radius is } \frac{1}{2} \times \frac{4 + \frac{7}{2}}{\sqrt{9+16}} = \frac{1}{2} \times \frac{15}{2} \times \frac{1}{5} = \frac{3}{4}$$

The centre of the circle lies on the line parallel to the given lines at a distance of $\frac{3}{4}$ from each of them. So let

$$\text{the equation be } 3x - 4y + k = 0 \quad \dots (i)$$

$$\text{For } k = \frac{1}{4} \text{ distance of (i) from the other line is } \frac{3}{4}.$$

$$\text{Thus the centre lies on the line } 12x - 16y + 1 = 0$$

3.(AC) $y = mx$

It makes an angle of $\pm 45^\circ$ with $2x + 3y = 6$

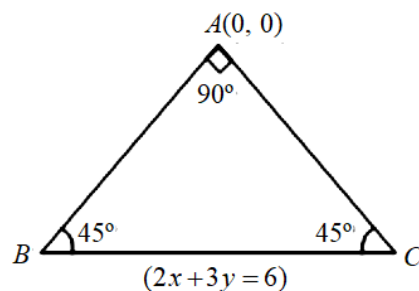
$$\tan(\pm 45^\circ) = \frac{m - \left(-\frac{2}{3}\right)}{1 + m\left(-\frac{2}{3}\right)} = \pm 1 \text{ or } 3m + 2 = \pm(3 - 2m) \Rightarrow m = \frac{1}{5}, -5$$

Hence sides are $x - 5y = 0$, $5x + y = 0$ and $2x + 3y = 6$

Solving in pairs, vertices are (0, 0)

$$\left(\frac{6}{13}, \frac{30}{13}\right), \left(\frac{30}{13}, -\frac{6}{13}\right)$$

$$\Delta = \frac{1}{2}(x_1y_2) - (x_2y_1) = \frac{1}{2} \times \frac{936}{169} = \frac{36}{13}$$



4.(AD) $S : (x-2)^2 + (y+3)^2 = 0$

$\therefore$ S is a point circle which represents point $(2, -3)$ and this point $(2, -3)$ also lies on the line $L : 2x + 5y + 11 = 0$

Equation of tangents from $(2, -3)$ to the circle

$$x^2 + y^2 = \frac{121}{29} \text{ is } y + 3 = m(x - 2) \Rightarrow mx - y = 3 + 2m$$

Applying $p = r$

$$\left| \frac{3+2m}{\sqrt{1+m^2}} \right| = \sqrt{\frac{121}{29}}$$

$$\Rightarrow 29(9 + 12m + 4m^2) = 121(1 + m^2) \Rightarrow 5m^2 - 348m - 140 = 0$$

$$\therefore m_1 + m_2 = \frac{348}{5} \text{ \& } m_1 m_2 = -\frac{140}{5} = -28$$

5.(ABCD) Let $f(x) = x^2 - 6x - a^2$

And as $(a, 0)$ lies on the diameter of the circle

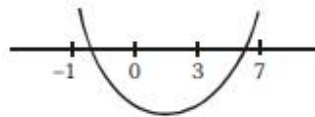
So maximum value of a^2 can be 4

Now $f(-1) = 1 + 6 - a^2 > 0$

$$f(0) = -a^2 < 0$$

$$f(3) = 9 - 18 - a^2 < 0$$

$$f(7) = 49 - 42 - a^2 > 0$$



So the equation $x^2 - 6x - a^2 = 0$ has one root between -1 and 0 & one root between 3 and 7 .

6.(AD) $g_1 = 2$ $g_2 = 0$

$$f_1 = m^2 + m \quad f_2 = m^2 + m$$

$$c_1 = 6 \quad c_2 = m^2 + m$$

$$2g_1g_2 + 2f_1f_2 = c_1 + c_2$$

$$\Rightarrow 2(m^2 + m)(m^2 + m) = (m^2 + m) + 6$$

$$m^2 + m = t$$

$$2t^2 - t - 6 = 0 \Rightarrow 2t^2 - 4t + 3t - 6 = 0$$

$$\Rightarrow 2t(t-2) + 3(t-2) = 0$$

$$\Rightarrow t = -\frac{3}{2} \quad t = 2$$

$$m^2 + m = -\frac{3}{2} \text{ (Not possible)}$$

$$m^2 + m = 2 \Rightarrow (m+2)(m-1) = 0$$

$$7.(A) \quad \because z^n - 1 = (z-1)(z-\alpha)(z-\alpha^2).....(z-\alpha^{n-1})$$

$$\Rightarrow \frac{z^n - 1}{z - 1} = (z - \alpha)(z - \alpha^2).....(z - \alpha^{n-1})$$

$$\Rightarrow \lim_{x \rightarrow 1} \frac{z^n - 1}{z - 1} = \lim_{x \rightarrow 1} (z - \alpha)(z - \alpha^2).....(z - \alpha^{n-1})$$

$$\Rightarrow n = \prod_{r=1}^{n-1} (1 - \alpha^r) = \prod_{r=1}^{n-1} (1 - \alpha^{\frac{2\pi r i}{n}})$$

$$= n = \prod_{r=1}^{n-1} \left\{ -e^{\frac{\pi r i}{n}} \left(e^{\frac{\pi r i}{n}} \right) - e^{-\frac{\pi r i}{n}} \right\} = \prod_{r=1}^{n-1} -e^{\frac{\pi r i}{n}} 2i \sin\left(\frac{r\pi}{n}\right)$$

$$\Rightarrow |n| = \prod_{r=1}^{n-1} 2 \sin\left(\frac{r\pi}{n}\right) \Rightarrow |n| = 2^{n-1} \prod_{r=1}^{n-1} \sin\left(\frac{r\pi}{n}\right) \Rightarrow \prod_{r=1}^{n-1} \sin\left(\frac{r\pi}{n}\right) = \frac{n}{2^{n-1}} \quad (\because n > 1)$$

$$\text{So } \sum_{r=1}^{n-1} \ln \sin \frac{r\pi}{n} = \ln \left(\prod_{r=1}^{n-1} \sin \frac{r\pi}{n} \right) = \ln \frac{n}{2^{n-1}} = \ln n - (n-1) \ln 2$$

$$8.(D) \quad n = 7, \alpha^7 = 1 \text{ and } 1 + \alpha + \alpha^2 + \dots + \alpha^6 = 0$$

$$\text{Sum of the roots} = \alpha + \alpha^2 + \alpha^4 + \alpha^3 + \alpha^5 + \alpha^6 = -1$$

$$\text{Product of the roots} = (\alpha + \alpha^2 + \alpha^4)(\alpha^3 + \alpha^5 + \alpha^6) = 2$$

$$\text{So the required equation is } x^2 + x + 2 = 0$$

$$9.(B) \quad \text{Image of A in } L_1 \text{ lies on BC}$$

$$\text{and image of A in } L_2 \text{ lies on BC}$$

$$A_1 \equiv (-2, -1)$$

$$\text{Let } A_2 \text{ be image of A in } L_2$$

$$A_2 \equiv (-1, 2)$$

$$\text{Slope of } A_1 A_2 = \text{Slope of BC}$$

$$10.(A) \quad \text{Image of A in } L_1 \text{ lies on BC and image of A in } L_2 \text{ lies on BC}$$

$$A_1 \equiv (-2, -1)$$

$$\text{Let } A_2 \text{ be image of A in } L_2$$

$$A_2 \equiv (-1, 2)$$

$$\text{Slope of } A_1 A_2 = \text{Slope of BC}$$

$$11.(A) \quad \text{Clearly } L_1 \text{ will pass through intersection point of } B_1 \text{ \& } B_2. \text{ So its equation will be}$$

$$(3x + 4y - 10) + \lambda(4x - 3y - 5) = 0 \text{ as } L_1 \text{ passes through origin so } \lambda = -2 \text{ hence equation of } L_1$$

$$-5x + 10y = 0$$

$$x = 2y$$

12.(B) As B_1 is equally inclined to L_1 & L_2 . Let slope of L_2 is m then

$$\frac{m - \frac{4}{3}}{1 + \frac{4m}{3}} = \frac{\frac{4}{3} - \frac{1}{2}}{1 + \frac{4}{3} \times \frac{1}{2}} \Rightarrow \frac{3m - 4}{3 + 4m} = \frac{1}{2} \Rightarrow 6m - 8 = 3 + 4m$$

$m = \frac{11}{2}$ also L_2 will pass through point of intersection of B_1 & B_2 i.e. (2, 1) so equation of L_2 is

$$y - 1 = \frac{11}{2}(x - 2)$$

$$2y - 2 = 11x - 22 \Rightarrow 11x - 2y = 20$$

13.(3) $(r+1)$ th term in the expansion of $(x^3 - 1/x^2)^n$ is ${}^nC_r (x^3)^{n-r} \left(-\frac{1}{x^2}\right)^r$
 $= {}^nC_r (-1)^r x^{3n-5r}$

For coefficient of x^5 , we set $3n - 5r = 5 \Rightarrow r = \frac{3n-5}{5} = p$ (say)

And for coefficient of x^{10} , we set $3n - 5r = 10 \Rightarrow r = \frac{3n-10}{5} = q$ (say)

$$p - q = 1$$

$$\text{Given, } {}^nC_p (-1)^p + {}^nC_q (-1)^q = 0 \Rightarrow n = 15.$$

$$\begin{aligned} 14.(9) \quad (1+3x+2x^2)^6 &= (1+x(3+2x))^6 \\ &= 1 + {}^6C_1 x(3+2x) + {}^6C_2 x^2(3+2x)^2 + {}^6C_3 x^3(3+2x)^3 + \dots + {}^6C_6 x^6(3+2x)^6 \end{aligned}$$

Now we get the term involving x^{11} from ${}^6C_6 x^6(3+2x)^6$ only.

$$\therefore \text{Coefficient of } x^{11} = {}^6C_5 3 \cdot 2^5 = 576 \quad \therefore k \times 2^6 = 9 \times 2^6 \quad \therefore k = 9$$

15.(3) Let $r > 0$ be the radius of the circle touching $x = 0$, $y = 0$, $3x + 4y = 12$ and whose centre lies in fourth quadrant. Then centre must be $(r, -r)$ (Since circle touches both axes). Since circle touches $3x + 4y - 12 = 0$

$$\left| \frac{3r - 4r - 12}{5} \right| = r \Rightarrow r = -2, r = 3$$

$$r = -2, \text{ is not possible} \Rightarrow r = 3$$

16.(5) Length of the perpendicular from the centre $(2, -1)$ on the tangent $3x + y = 0$ is

$$\left| \frac{6-1}{\sqrt{9+1}} \right| = \frac{5}{\sqrt{10}},$$

Which is the radius of the circle. Equation of the circle is

$$(x-2)^2 + (y+1)^2 = \frac{5}{2} \quad \text{or} \quad [(x-2)^2 + (y+1)^2] - 5 = 0$$

$$\text{The required length} = [(23-2)^2 + (17+1)^2] - \frac{5}{2}$$

17.(7) Let (h, k) be any point on the first circle then $h^2 + k^2 + 70h - 45k + 6789 = 0$

Square of the length of the tangent from (h, k) to the second circle is

$$\begin{aligned} h^2 + k^2 + 70h - 45k + 9876 \\ = -6789 + 9876 = 3087 \end{aligned}$$

18.(8) $2x - y + 1 = 0$ is tangent

Slope of line $OA = -\frac{1}{2}$, equation $(y - 5) = -\frac{1}{2}(x - 2)$

$$2y - 10 = -x + 2$$

$$x + 2y = 12$$

Intersection with $x - 2y = 4$ will give coordinates of centre

$$x + 2y = 12$$

$$x - 2y = 4$$

$$\begin{array}{r} - \quad + \\ \hline \end{array}$$

$$4y = 8 \quad \Rightarrow \quad y = 2$$

$$x - 4 = 4 \quad \Rightarrow \quad x = 8$$

$$C = (8, 2)$$

$$\text{Distance } OA = \sqrt{(8-2)^2 + (2-5)^2} = \sqrt{36+9} = \sqrt{45} = 3\sqrt{5}$$

